

ECS Project Step 7 – ECS Service with Application Load Balancer

To Begin with the Lab

Summary of the Lab

In this lab, the process of creating an **Amazon ECS Service** was explained as the way to host the application in **Amazon ECS** with high availability and load balancing. The video showed why the service should use a task definition, desired task count, private subnets for the tasks, and an **Application Load Balancer** in the public subnet. It also explained that the load balancer should be created separately, along with a target group, so the correct subnet can be chosen and the architecture follows AWS best practices. The service was intended to keep tasks running, distribute traffic through the load balancer, and register ECS tasks automatically in the target group. The session ended by preparing for autoscaling and a full service-based deployment in the next step.

Understanding Amazon ECS Service and Load Balancing

An **Amazon ECS Service** keeps the required number of tasks running and helps make an application available all the time. It exists because a single task is temporary, but a service can restart tasks and work with a load balancer to distribute traffic across them. In this lab, the service is used with an **Application Load Balancer** so users can reach the app from the internet while the ECS tasks stay inside private subnets. For example, if one task stops, the service starts another one, and the load balancer keeps sending requests to the healthy tasks, which gives better availability and a smoother user experience.

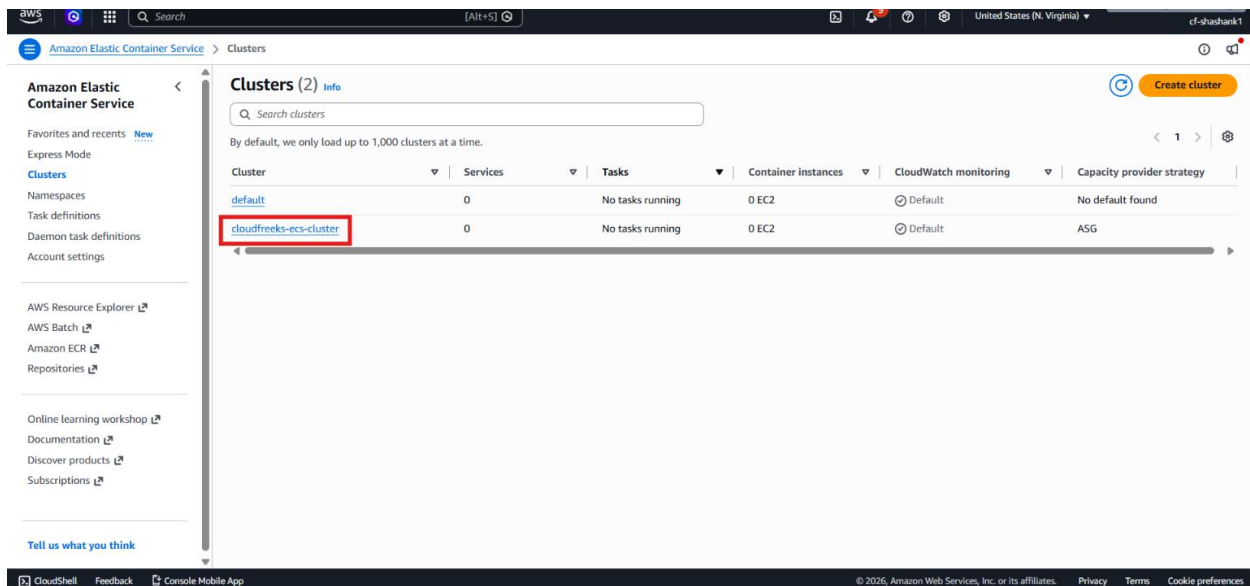
Example:

A PHP web app is deployed with two ECS tasks so it can stay available even if one task fails. The **Application Load Balancer** sits in the public subnet and forwards web traffic to the ECS tasks running in private subnets. If one task becomes unhealthy, ECS replaces it automatically and the load balancer keeps the app accessible.

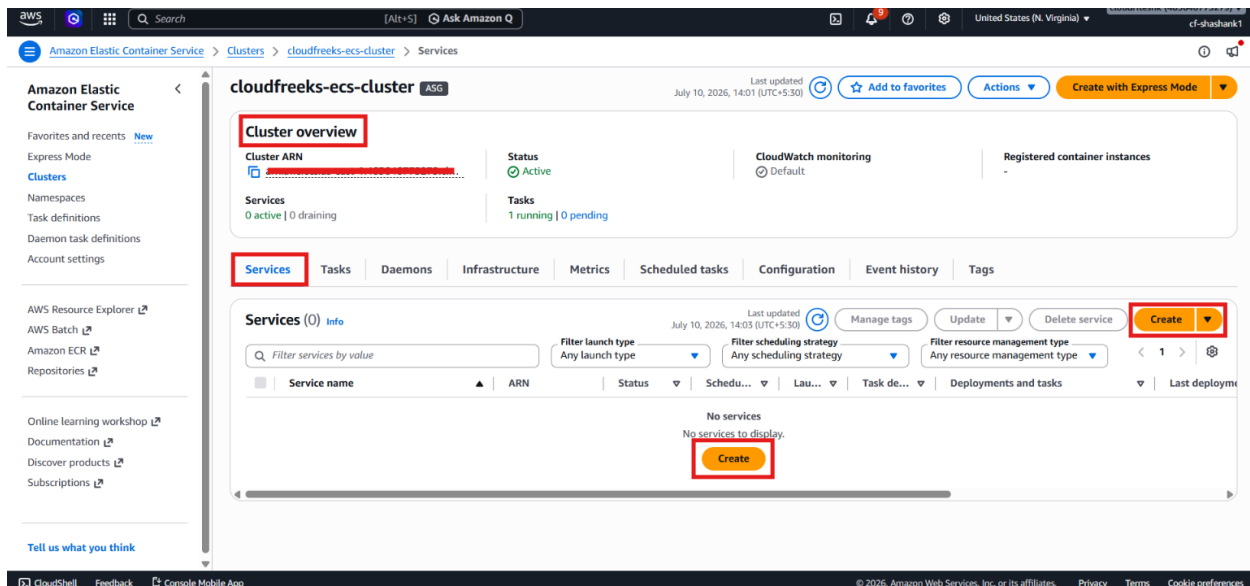
Lab Steps

Step 1 – Open the Service Creation Page

- Sign in to the **AWS Management Console**.
- Open **Amazon ECS**.
- Select the existing cluster.

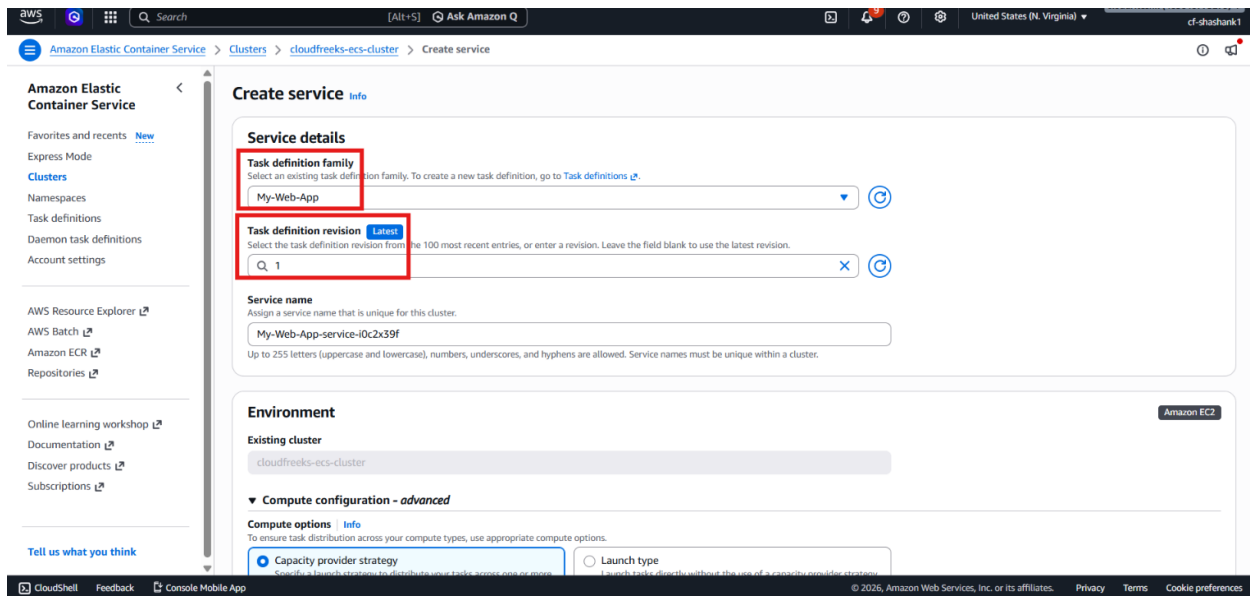


- Click **Create service**.

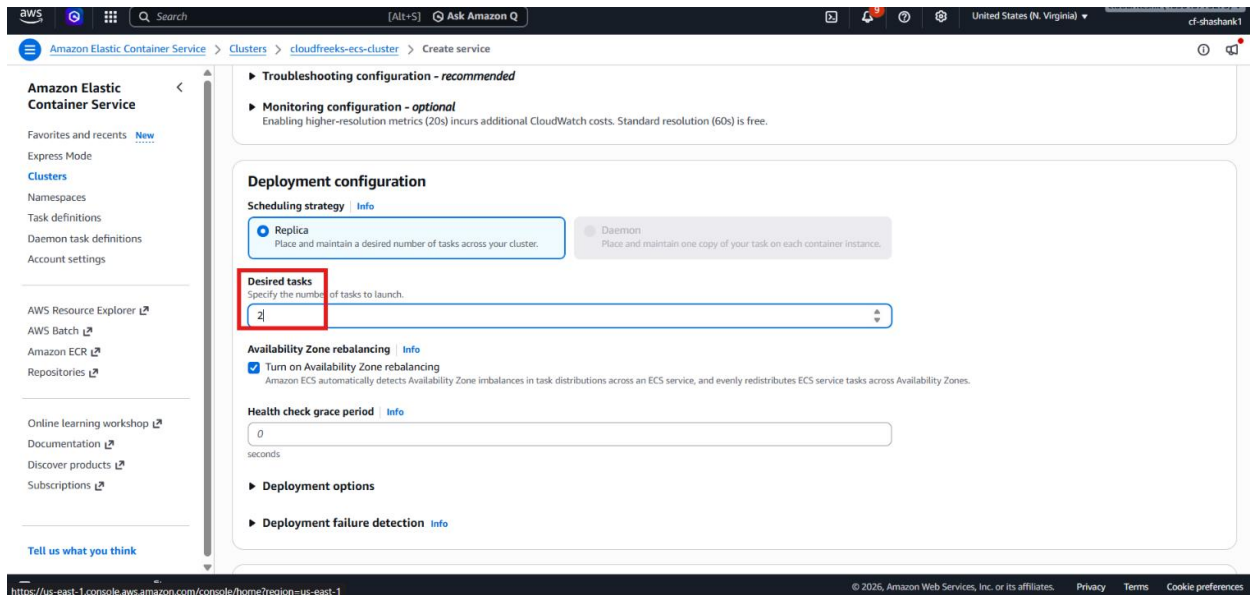


Step 2 – Select the Task Definition

- Choose the task definition created in the previous lab.
- Select the correct task definition revision.



- Set the desired number of tasks to 2.



Step 3 – Configure Networking

- Choose the VPC that contains the private and public subnets.
- Select only the private subnets for the ECS tasks.
- Use the default or existing security group for the service.

Amazon Elastic Container Service

Clusters > cloudfreeks-ecs-cluster > Create service

Networking

VPC | Info
Select a VPC to use for your Amazon ECS resources.
vpc-0f652401038eb2a22
My VPC

Subnets
Choose the subnets within the VPC that the task scheduler should consider for placement.
Choose subnets
subnet-0f5c2d6baac91163f
Private Subnet_2
subnet-05270892ff67cb83a
Private Subnet_1

Security group | Info
Choose an existing security group or create a new security group.
☒ Use an existing security group
☐ Create a new security group

Security group name
Choose an existing security group.
Choose security groups
sg-01a6920f81b85f995
default

Service Connect - optional | Info
Service Connect allows for service-to-service communications with automatic discovery using short names and standard ports.

- Turn Off the **Public IP** if enabled.
- Keep the tasks away from the public subnet so they are not exposed directly to the internet.

Step 4 – Prepare Load Balancing

- Enable load balancing.
- Select **Application Load Balancer**.
- Set the container port to 80.

Amazon Elastic Container Service

Clusters > cloudfreeks-ecs-cluster > Create service

Load balancing - optional | Info
Configure load balancing using Amazon Elastic Load Balancing to distribute traffic evenly across the healthy tasks in your service.

☒ Use load balancing

VPC
The VPC for your load balancing resources must be the same as the VPC for your service with awsvpc.
vpc-0f652401038eb2a22

Load balancer type | Info
Specify the load balancer type to distribute incoming traffic across the tasks running in your service.
☒ Application Load Balancer
Application Load Balancer makes routing decisions at the application layer (HTTP/HTTPS), supports path-based routing, and can route requests to one or more ports.
☐ Network Load Balancer
A Network Load Balancer makes routing decisions at the transport layer (TCP/UDP).

Container
The container port to load balance the incoming traffic to
web 80:80
Host port:Container port

Application Load Balancer
Specify whether to create a new load balancer or choose an existing one.
☒ Create a new load balancer
☐ Use an existing load balancer

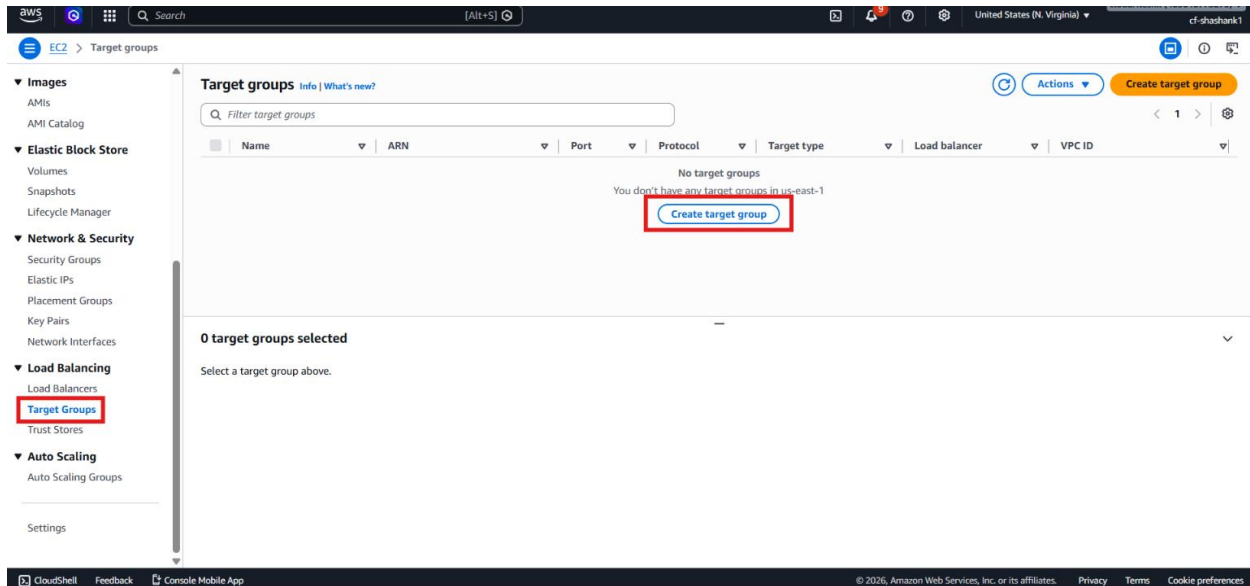
Load balancer name
Assign a unique name for the load balancer.
Creates an internet-facing Application Load Balancer. To create an internal load balancer, use the Amazon EC2 console.

- Avoid creating the load balancer directly from the ECS service page because subnet selection is limited there.

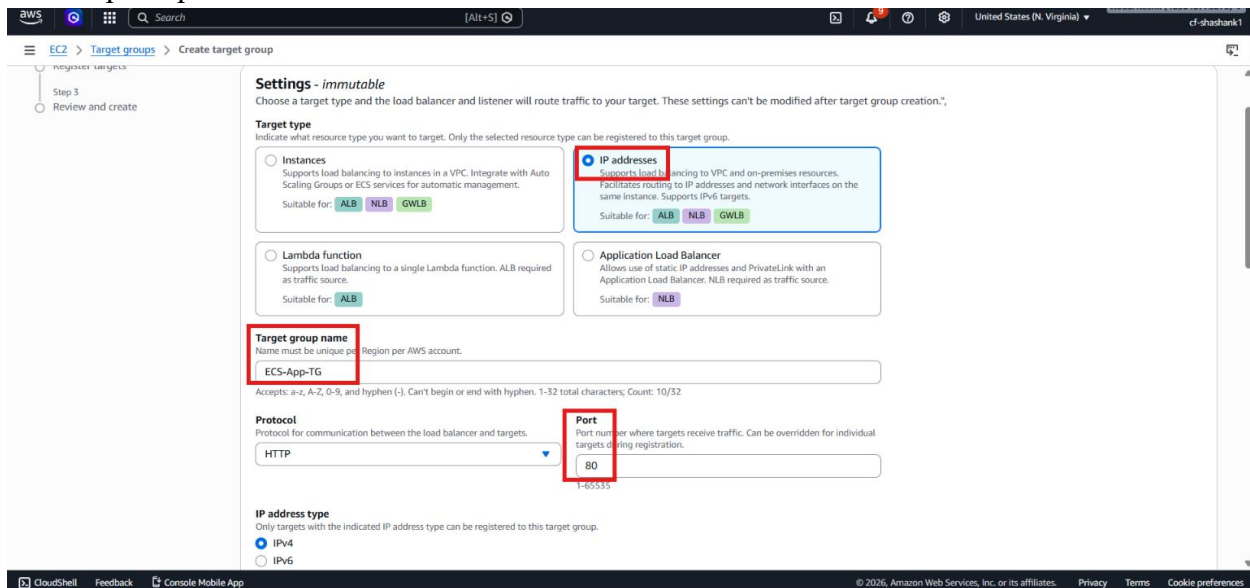
Step 5 – Create the Target Group First

- Open EC2.

- Go to **Target Groups**.
- Click **Create target group**.



- Select **IP addresses** as the target type.
- Enter the target group name.
- Keep the port as 80.



- Choose the same VPC.

IP address type
Only targets with the indicated IP address type can be registered to this target group.

☒ IPv4
☐ IPv6

VPC
Select the VPC that hosts the load balancer. Only VPCs that support the IP address type selected above are available in this list. On the [Register targets](#) page, you can register IP addresses from this VPC, or from private IP addresses located outside of this load balancer's VPC (such as a peered VPC, EC2-Classic, or on-premises targets that are reachable over Direct Connect or VPN).

vpc-0f652401038eb2a22 (My-VPC)
192.168.0.0/24

[Create VPC](#)

Protocol version

☒ HTTP1
Send requests to targets using HTTP/1.1. Supported when the request protocol is HTTP/1.1 or HTTP/2.

☐ HTTP2
Send requests to targets using HTTP/2. Supported when the request protocol is HTTP/2 or gRPC, but gRPC-specific features are not available.

☐ gRPC
Send requests to targets using gRPC. Supported when the request protocol is gRPC.

Health checks
The associated load balancer periodically sends requests, per the settings below, to the registered targets to test their status.

Health check protocol
HTTP

Health check path
Use the default path of "/" to perform health checks on the root, or specify a custom path if preferred.
/

- Create the target group.

Step 6 – Create the Application Load Balancer

- Open EC2.
- Go to **Load Balancers**.
- Click **Create load balancer**.

Load balancers [What's new?](#)

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

Filter load balancers

Name	State	Type	Scheme	IP address type	VPC ID	Availability Zones	Security group
0 load balancers selected							

Select a load balancer above.

- Choose **Application Load Balancer**.

Load balancer types

Application Load Balancer

Choose an Application Load Balancer when you need a flexible feature set for your applications with HTTP and HTTPS traffic. Operating at the request level, Application Load Balancers provide advanced routing and visibility features targeted at application architectures, including microservices and containers.

Create

Network Load Balancer

Choose a Network Load Balancer when you need ultra-high performance, TLS offloading at scale, centralized certificate deployment, support for UDP, and static IP addresses for your applications. Operating at the connection level, Network Load Balancers are capable of handling millions of requests per second securely while maintaining ultra-low latencies.

Create

Gateway Load Balancer

Choose a Gateway Load Balancer when you need to deploy and manage a fleet of third-party virtual appliances that support GENEVE. These appliances enable you to improve security, compliance, and policy controls.

Create

- Enter the load balancer name.
- Select **Internet-facing**.
- Choose the same VPC.
- Select the public subnets.

Create Application Load Balancer

Network mapping

The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC

The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view [target groups](#).

vpc-0f652401038eb2a22 (My-VPC)
192.168.0.0/24

Create VPC

IP pools

You can optionally choose to configure an IPAM pool as the preferred source for your load balancers IP addresses. Create or view Pools in the [Amazon VPC IP Address Manager console](#).

☐ Use IPAM pool for public IPv4 addresses

The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.

Availability Zones and subnets

Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

☒ us-east-1a (use1-az4)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL inbound and outbound ALLOW rules, and internet gateway (IGW) routes configured to allow traffic.

subnet-0e4a9f94c1225c2d (Public_Subnet_1)
192.168.0.0/26
Network ACL ALLOW rules: Inbound (1) Outbound (1) | Route table: IGW (missing)

Reachability may be impacted

☒ us-east-1b (use1-az6)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL inbound and outbound ALLOW rules, and internet gateway (IGW) routes configured to allow traffic.

subnet-0aa878cb01684a4f (Public_Subnet_2)
192.168.0.64/26
Network ACL ALLOW rules: Inbound (1) Outbound (1) | Route table: IGW (missing)

Reachability may be impacted

- Attach the default security group or an existing security group.
- Select the existing target group.

Listeners and routing [info](#)

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

Listener HTTP:80 [Remove](#)

Protocol: HTTP Port: 80

Default action [info](#)

The default action is used if no other rules apply. Choose the default action for traffic on this listener.

Routing action

☒ Forward to target groups ☐ Redirect to URL ☐ Return fixed response

Forward to target group [info](#)

Choose a target group and specify routing weight or [create target group](#).

Target group: ECS-App-TG HTTP Weight: 1 Percent: 100%

Target group stickiness [info](#)

Enables the load balancer to bind a user's session to a specific target group. To use stickiness the client must support cookies. If you want to bind a user's session to a specific target, turn on the Target Group attribute Stickiness.

☐ Turn on target group stickiness

Listener tags - optional

Consider adding tags to your listener. Tags enable you to categorize your AWS resources so you can more easily manage them.

- Create the load balancer.

Step 7 – Return to ECS Service Creation

- Return to **Amazon ECS**.
- Open **Create service** again if needed.
- Choose the same task definition.
- Set the desired task count.
- Select the private subnets.
- Enable **Application Load Balancer**.
- Choose the existing load balancer.

Amazon Elastic Container Service

Load balancing - optional

Configure load balancing using Amazon Elastic Load Balancing to distribute traffic evenly across the healthy tasks in your service.

☒ Use load balancing

VPC

The VPC for your load balancing resources must be the same as the VPC for your service with awsvpc.

vpc-0f652401038eb2a22

Load balancer type [info](#)

Specify the load balancer type to distribute incoming traffic across the tasks running in your service.

☒ Application Load Balancer ☐ Network Load Balancer

Container

The container and port to load balance the incoming traffic to.

web 80:80

Application Load Balancer

Specify whether to create a new load balancer or choose an existing one.

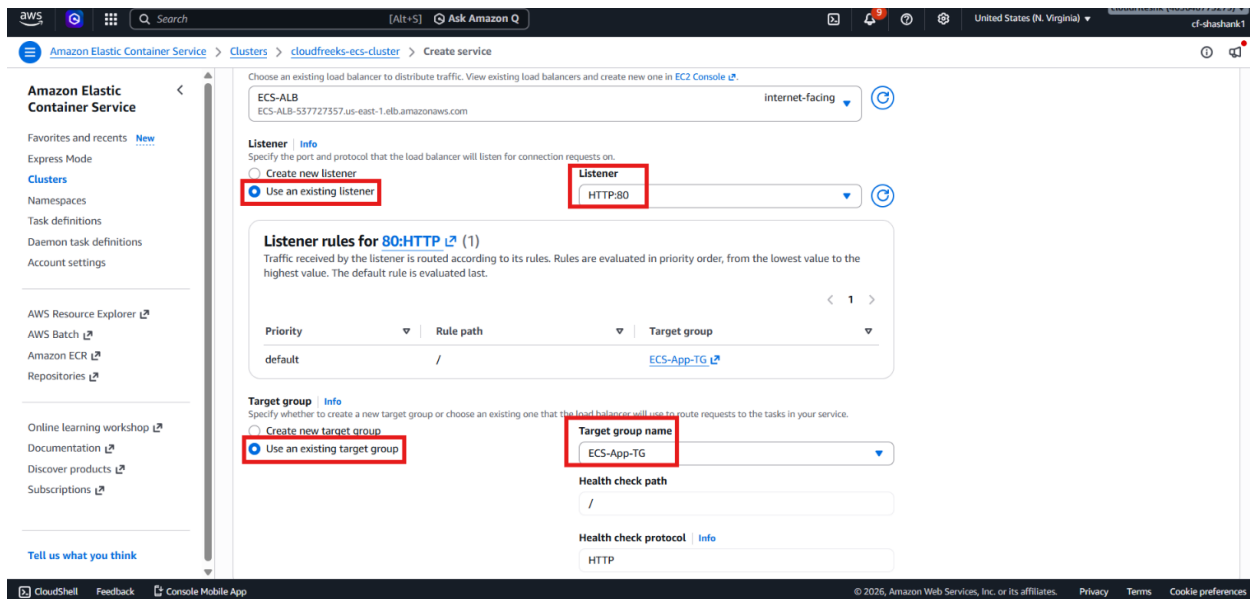
☐ Create a new load balancer ☒ Use an existing load balancer

Load balancer

Choose an existing load balancer to distribute traffic. View existing load balancers and create new one in [EC2 console](#).

ECS-ALB

- Choose the existing listener.
- Choose the existing target group.



- Create the service.

Verification

- Verified that the ECS service was created using the existing task definition.
- Verified that the desired number of tasks was set correctly.
- Verified that ECS tasks were placed in private subnets.
- Verified that an **Application Load Balancer** was created in the public subnet.
- Verified that a target group using **IP addresses** was created successfully.
- Verified that the load balancer and target group were attached to the ECS service.
- Verified that ECS tasks can be registered automatically with the load balancer for traffic routing.